ReScene: A Hybrid Semantic and Structural Framework for Revisit-Based Video Forgery Detection

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***Abstract*—The easy availability of AI-enabled video editing software has made it quite simple to create convincing forgeries of videos which have raised serious questions on the authenticity of visual evidences. While current video forensic methods mainly rely on identifying pixel level artifacts and inconsistencies in temporal domain, they are not able to establish whether the semantic information of a revisited scene is consistent in a video or not. In this work, we present ReScene, a hybrid video forgery detection system which integrates semantic scene understanding and structure verification for revisit scene-based video analysis. First, the proposed framework performs object detection using YOLOv8 and generates scene graph of detected entities showing the contextual relationships among entities. Scene Memory Module accumulates the object appearance and relationship statistics along the whole video which can be utilized to discover semantic inconsistencies via Graph Convolutional Network due to object insertion, deletion, replacement or unnatural interaction between objects. Simultaneously, an independent structure verification module leverages Structural Similarity Index (SSSIM) and ORB feature matching to detect local visual alterations in continuous scene parts. Finally, the outputs of both branches are adaptively combined to generate the tampering prediction and provide evidence for the tampering verdict.**

***Index Terms*—Video Forgery Detection, Scene Graphs, Graph Convolutional Networks, SSIM, ORB Feature Matching, Digital Forensics**

I. Introduction

The fast development of AI, deep learning, and multimedia editing techniques has greatly affected the process of producing and distributing videos online. With the help of modern editing tools and machine learning-based models, forgery of videos has become more realistic through the introduction of objects, erasure of objects, object replacement, and face modification. Although this technology has sped up the process of creating content, it has also created serious threats to the credibility of digital media in terms of its usage in surveillance, journalism, criminal investigation, insurance claims, and social networks [1], [2].

In contrast to conventional methods of image tampering, video forgery is concerned with spatio-temporal consistency, thus making its detection much more difficult. Video manipulations that occur in a series of frames must ensure visual consistency without the creation of any artifacts in order to make it impossible for sophisticated forgeries to be noticed. This led to the development of the field of passive digital video forensics, which focuses on finding out the authenticity of multimedia data independently of any embedded signatures or watermarking. [3], [4].

The recent developments in the field of deep learning have significantly e [5].hanced the video forensics techniques by using low-level visual artifacts such as inconsistencies in video compression, sensor pattern noise, color interpolation artifacts, and generative models' fingerprint [1]. CNNs and transformers have shown excellent results in detecting deepfakes and synthetic media; however, their limitations have been that these approaches mainly concentrate on the detection of pixel-level manipulations artifacts of an individual frame. While being highly useful in detecting digital alterations, they fail to reason about the temporal consistency of the semantics of a physical scene in the video [5].

A large number of applications in practice, such as surveillance, security auditing, inventory checking, and self-inspection, necessitate the detection of any unauthorized modifications of an earlier inspected environment. In these revisit scenarios, the task is not just about detecting visual anomalies; rather, it is about determining if there has been any addition, deletion, movement, or substitution of objects between the different observations of the same scene. This particular task has seldom been addressed using pixel-based methods since both semantic and structural consistency need to be taken into

account.

The study of scenes is now gaining importance in the field of computer vision as it allows reasoning over a scene as well as beyond just recognizing objects in the scene. Scene graphs capture the semantics as well as the spatial relation between various objects in the scene, and hence allow modeling of complicated interactions taking place in a scene. When combined with GCNs, scene graphs perform well on problems related to relational reasoning and visual understanding [6]–[8].

While all of the above is true, current scene graph techniques are inherently limited by the performance of the object detector used to detect objects within the scenes. This is because object detectors are designed to recognize only predefined sets of object classes, and therefore they are unable to recognize changes in the structures containing objects unknown to the detectors' vocabulary. Also, minor changes in scenes like removing an object, moving it, or modifying the background can keep the same object labels but still greatly change the overall scene structure.

In order to overcome these problems, this paper introduces a novel hybrid video tampering detection approach called **ReScene**. The first pathway is responsible for object detection in video frames, creation of graph structures denoting the relationship between objects, and application of Graph Convolutional Network for detecting inconsistency in time and relation of objects based on a video forgery benchmark dataset. On the other hand, the second pathway analyzes structural changes in revisited scenes using Structural Similarity Index (SSIM), Regional Difference Analysis, and ORB feature matching. [9], [10].

With the help of combining relational scene reasoning and pixel-level structural verification, the proposed framework seeks to achieve a more complete and interpretable solution for video tampering detection than the solutions that make use of only one of the two detection pathways.

The rest of the paper is organized as follows. In Section II, we briefly review the existing literature related to video forgery detection and graph-based visual reasoning. The ReScene framework and both semantic and structural detection pathways are described in Section III. Implementation and experimental settings are discussed in Section IV. Experimental results are presented in Section V. Limitations of the proposed method are outlined in Section VI. Finally, the conclusions of this work are stated in Section VII.

II. Related Work

A. Video Forgery Detection

With the growing number of advanced video editing tools and the development of generative artificial intelligence, studies on the identification of passive video manipulation have been increasingly explored. In contrast to active methods of verification that use digital signatures or watermarking for media validation, which are incorporated into the content when it is acquired, passive forensic techniques detect forgery through statistical traces present in the video during image acquisition and processing stages [1], [3], [11].

Initial forensics techniques made use of hand-engineered features which were created using the pipeline of image capture by camera. Photo Response Non Uniformity (PRNU), sensor noise, temporal noise correlations, inconsistent compression, color filter array interpolation errors, and motion discontinuity have been extensively used for detecting copy move forgery, frame insertion, frame deletion, temporal duplication, and object removal [4], [12]–[14]. While these methods yielded promising results in an ideal setting, they are highly prone to robust editing methods including strong compression and post processing [1]

Deep learning techniques have achieved significant progress in video forgery detection due to the learning of discriminative visual representations from large datasets. CNNs, attention models, and transformers have proved to be very effective for deepfake and synthetic media detection based on their ability to analyze appearance inconsistencies, temporal inconsistencies, and frequency domain features [5], [15], [16]. However, many existing approaches are mainly concerned with appearance-based evidence from individual frames without capturing higher-order semantic inconsistencies among the objects present in the scene.

A qualitative analysis of representative video forgery detection approaches is depicted in Table I. It becomes apparent from the table that most of the techniques available either rely on structure-based validation or semantic understanding. This leads to the fact that no unified approach exists which incorporates both sources of information.

TABLE I
Comparison of existing video forgery detection methods.

Method Category	Semantic	Structural	Primary Limitation
Noise Residue / PRNU	×	✓	Compression sensitive
CNN-based Detection	×	✓	Appearance focused
Transformer-based Detection	Limited	✓	Large training data
Scene Graph Learning	✓	×	Detector dependent

B. Semantic Scene Understanding

The ability of scene understanding has proved to be crucial in computer vision as it allows machines to think beyond recognizing individual objects. Scene graphs capture visual scenes through graph structures, wherein objects are nodes and semantic/spatial relations are graph edges. The structure thus captures the context in which the interaction takes place between entities in the scene, resulting in better visual relation detection, image retrieval, image captioning, and visual question answering when compared to traditional approaches [6], [7], [17], [18].

The Graph Neural Networks (GNNs) and specifically Graph Convolutional Networks (GCNs) improve upon scene graph representations by enabling the propagation of context information among neighboring nodes for learning the semantic dependencies which cannot be learned from individual objects alone [8], [19]. However, despite their success in learning

semantic dependencies, graph-based approaches have not been widely utilized in multimedia forensics due to the fact that existing approaches are mainly concerned with improving the understanding of scenes and not detecting any contradictions that may arise due to the malicious modification of the scenes. Moreover, it is important to note that the creation of a scene graph requires good detection of objects which may not be possible for unseen objects.

C. Research Gap

Despite significant advancements made in the realm of passive multimedia forensics and semantic scene understanding, the current literature shows that there is still room for improvement. While appearance-based forensic approaches can efficiently detect low-level manipulations in the image, graph-based approaches are effective at modeling contextual relations between the scene elements. Still, both approaches do not work together and cannot take advantage of the semantic reasoning and structural verification in combination with each other. This is why manipulations that change either the visual appearance or the semantic context without changing the other one become hardly detectable. Therefore, the developed **ReScene** framework combines semantic scene graph-based reasoning with SSIM and ORB features matching.

III. Proposed Methodology

A. Overall Framework

The proposed **ReScene** model detects tampered videos based on a joint analysis of semantic and visual consistency of scenes using two separate processing streams, which is depicted in Fig. 2. As opposed to the traditional forensic models, which consider either appearance-related or semantic information only, the proposed framework utilizes both types of evidence.

Consider the case of an input reference video V_r and query video V_q. The extraction of representative frames from these two videos is done at fixed time intervals and results in a set of frames denoted by

$$F = \{f_1, f_2, \ldots, f_N\}, \tag{1}$$

where N is the total number of frames extracted.

This set of frames is processed using two parallel pathways. In the semantic pathway, YOLOv8 is used to detect objects present in scenes and generate scene graphs that depict the spatial relationships between these objects. The *Scene Memory* module keeps track of the object appearances, relationship frequencies, and contextual co-occurrence across the entire reference video to facilitate semantic reasoning for longer periods of time.

At the same time, the structural verification path works directly on the extracted frames without making use of object detection. The stable frames from the reference video and the query video are then used in SSIM and ORB feature matching to detect any inconsistencies in terms of structural information due to object deletion or insertion.

Lastly, the two types of confidence scores are combined using an adaptive decision fusion approach to get the tampering confidence score. The forensic module is then used to identify the suspicious elements, relationships, and structural abnormalities responsible for making the final prediction. With its combination of semantic reasoning and pixel-wise verification, ReScene makes a reliable framework for hybrid video forgery detection.

B. Scene Graph Construction and Semantic Analysis

Semantic approach represents each video frame as a structured graph which captures the context of the objects in a particular frame through the relations between them. The major difference between the conventional appearance-based approaches where objects are analyzed independently and the new one lies in the interaction representation.

Objects in each extracted frame are detected using YOLOv8 [20]. Every detected object represents a node of a graph, while the spatial relations between objects such as *left-of*, *right-of*, *above*, *below*, and *near* are used for graph edges representation.

$$G = (V, E), \tag{2}$$

where V stands for the detected objects, and E means the relationships between them.

Figure 1 demonstrates the Scene Memory derived from the reference video. The memory graph holds the relationship between objects that are observed regularly, and the co-occurrences keep track of the long-term semantics of the multiple frames. These semantics are then utilized by the MemoryGCN in detecting inconsistencies in the revisits scenes.

Then, such a constructed graph is applied to Graph Convolutional Network (GCN) with two layers [8]. With node feature matrix X and adjacency matrix A, graph convolution can be represented by

$$H^{(l+1)} = \sigma\left(\hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right), \tag{3}$$

in which $\hat{A} = A + I$ is the adjacency matrix with self-loops, $\hat{D}$ is its associated degree matrix, $W^{(l)}$ is the trainable weight matrix, and $\sigma(\cdot)$ is the activation function.

Because there is no such large dataset of semantic forgery, the GCN model is trained using artificial forged scenes graph based on object replacement, relation modification, random addition, and edge deletion.

$$S_{\text{sem}} \in [0, 1], \tag{4}$$

The larger the score, the higher the probability of being semantically inconsistent. Then, the confidence score is delivered to the adaptive fusion module.

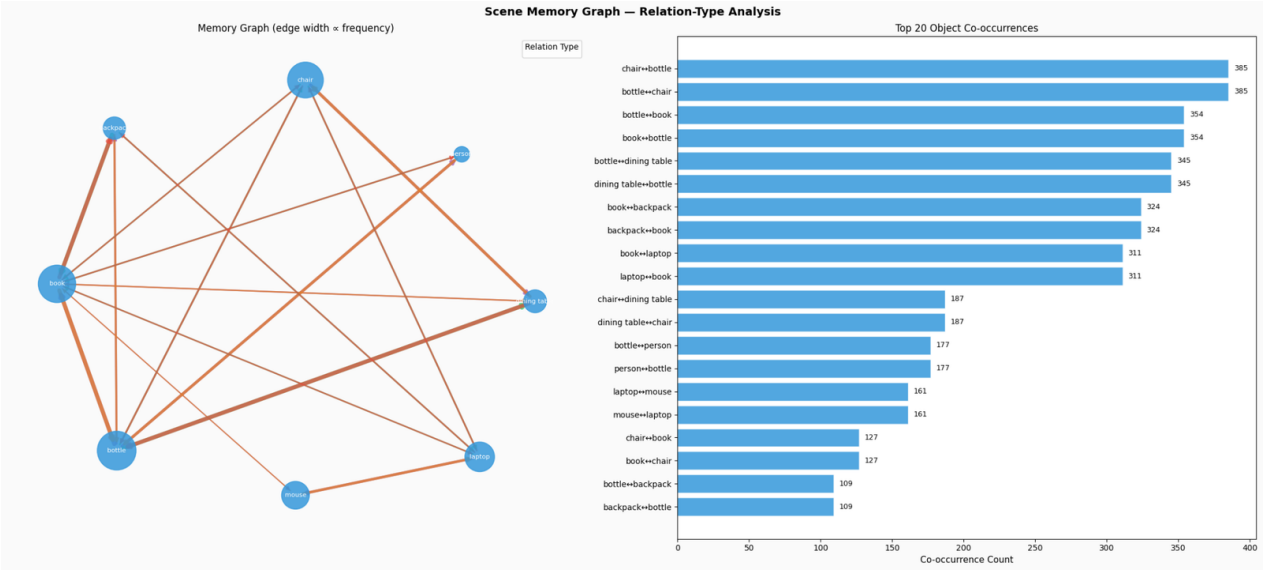


Fig. 1. Illustration of Scene Memory representation that would be employed in the proposed ReScene system. Left part shows the graph that was learned in terms of objects' relationships; note that the size of the node stands for the frequency of objects and edge's width shows relation frequency. Right part highlights most common object pairs learned from reference video.

C. Structural Verification

Although the semantics branch is able to detect contextual inconsistencies, some manipulations keep intact the relationships between objects, while changing the visual appearance. In order to tackle the problem, ReScene includes a separate structural verification branch processing the image pixels independently.

Firstly, the stable frames are chosen from both reference and query video sequences. For each pair of the corresponding frames, the Structural Similarity Index (SSIM) is calculated as

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \quad (5)$$

where μ, σ, and σ_{xy} are the local means, variances, and covariances of the compared frames correspondingly [9]. The regions that have low structural similarity are further verified with the help of ORB

The structure analysis outputs the normalized confidence score

$$S_{\text{struct}} \in [0,1], \quad (6)$$

with higher scores meaning the evidence of structure inconsistency is stronger. This confidence score is then used in combination with the semantic confidence score via the adaptive fusion method.

D. Adaptive Confidence Fusion

The two approaches offer supportive findings on tampering within videos. Where the semantic approach highlights any inconsistencies between the entities within the scene, the structural approach identifies any modifications in the pixel level. This is as shown in Fig. 2, where the output from the two approaches is adaptively fused to determine the outcome.

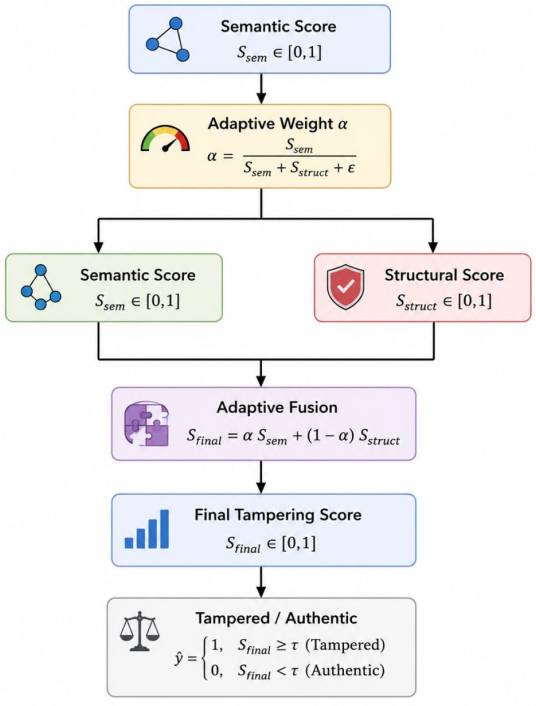


Fig. 2. Adaptive confidence fusion approach combining semantic and structural confidence measures to make the final prediction for video manipulation.

Let S_{sem} and S_{struct} be the normalized semantic and structural confidence scores, respectively. The tampering confidence score can be calculated using the following formula:

$$S_{\text{final}} = \alpha S_{\text{sem}} + (1-\alpha) S_{\text{struct}}, \quad (7)$$

where $\alpha \in [0, 1]$ is the adaptive fusion weight,

$$\alpha = \frac{S_{\text{sem}}}{S_{\text{sem}} + S_{\text{struct}} + \epsilon}, \tag{8}$$

and ϵ is a small value to avoid division by zero.

The final prediction is made based on the comparison of the obtained confidence

IV. Experimental Setup

A. Dataset and Benchmark Construction

As there is a lack of publicly available data sets for semantic video manipulations, a synthetic data set was created based on the authentic video. The authentic graphs representing the relationships between the detected objects were obtained by processing the representative frames of the video through the proposed scene graph generation pipeline.

The fake samples were generated by applying the Semantic Graph Perturbation Module to the authentic graphs. The following five techniques for creating the inconsistencies in the context were considered: object replacement, relation corruption, random object addition, edge removal, and spatial perturbation. The techniques provide the ability to create realistic forgeries without additional manual labeling.

The created data set consists of the same amount of authentic and fake samples to prevent any kind of class imbalance in the training process. In the end, the obtained data set was shuffled and split into training and testing sets by a ratio of 80:20 using the stratified sampling approach.

B. Semantic Model Training and Evaluation Dataset

In the procedure of the introduced ReScene approach, there are two phases of experiment analysis: training of the semantic model and assessment of the entire ReScene framework.

In order to train the semantic detector, the frames that represent the actual reference video have been transformed into scene graphs that provide information about the objects and their relations.

Due to the absence of publicly available datasets containing scene graph forgeries, the samples of forgery were generated using the Semantic Graph Perturbation Module. The following five different corruption procedures were applied: object substitution, relation corruption, random object addition, edge deletion, and spatial perturbation. In this way, the inconsistency in context is created for the forgery, preserving the graph structure and thus training on the graph forgery samples becomes possible without any annotation.

The entire ReScene method has been tested separately on the new dataset with revisit videos. It includes pairs of scenes, in which each pair consists of the video shot in the reference mode and true and forged revisit videos taken from the same angle. The forged videos contain object manipulation, such as object deletion, addition, replacement, and movement. The test included providing all the videos to the approach at once.

TABLE II
Case Study Analysis of ReScene

Case	Manipulation	Semantic Evidence	Structural Evidence
1	Object addition/removal	Object/relation change	Local structural change
2	Object displacement	Spatial relation change	Feature mismatch
3	Unknown/altered object	Limited detector evidence	SSIM/ORB inconsistency

C. Implementation Details

The suggested ReScene method was implemented using Python language and PyTorch, PyTorch Geometric libraries. For object detection and scene graphs creation, YOLOv8n [20] was used. For frames extraction, stable frames selection, ORB features extraction, and structural similarity verification, OpenCV library was used. Graph neural network models were implemented using the Graph Convolutional Network (GCN) architecture from the PyTorch Geometric library.

MemoryGCN semantic detector was trained using the Adam optimizer with the learning rate equal to 1×10^{-3} and the batch size of 32. Dropout regularization and early stopping were applied to prevent overfitting, while the best model in terms of performance on the testing set was selected for inference. For tampering assessment during inference, the adaptive confidence fusion technique for combining the semantic and structural confidence scores was applied.

V. Results and Discussion

A. Case Study Analysis

The practical behavior of ReScene was assessed using revisit scenarios based on the kind of scene alteration and the evidence generated by the two approaches. These scenarios are discussed in Table II.

In case of addition or deletion of an object, the semantic pathway detects changes in the semantics of entities and relations, whereas SSIM and ORB offer evidence based on localized structure. If a particular object gets relocated, its semantic category will not change, yet the new relationships will be detectable using the MemoryGCN algorithm, whereas additional evidence for that will come from the structural pathway based on feature mismatch. If the object is one of those that are not well-represented in the vocabulary used by the detector, there might not be enough semantic evidence; in such scenarios, however, structural pathway that does not depend on the detector will detect visual anomalies.

B. Quantitative Performance Evaluation

The presented ReScene model has been evaluated for the custom revisited video dataset, which comprises real and synthetic revisit scenes. The semantic MemoryGCN detector and the structural verification branch separately analyze the scene before the results are combined together using the proposed adaptive confidence fusion method. The output result is compared to the corresponding ground-truth label to determine the performance of the model.

The performance metrics of the presented ReScene model are presented in Table III.

Based on the experiment results, it can be concluded that the semantic and structural branches complement each other

TABLE III
PERFORMANCE OF RESCENE ON THE CUSTOM REVISITED-SCENE DATASET.

Method	Acc.	Prec.	Recall	F1
ReScene	0.77	0.78	0.75	0.76

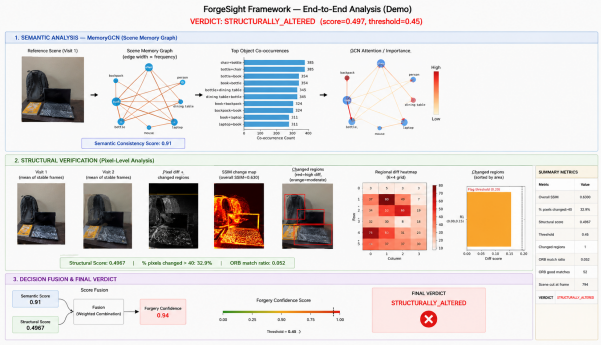


Fig. 3. Representative qualitative results of the proposed ReScene framework showing authentic and forged revisit scenes together with the corresponding semantic and structural analysis.

during the inference. The MemoryGCN detector determines the contextual inconsistencies associated with the insertion, deletion, replacement, and motion of objects, while the structural branch finds localized changes in visual scenes based on the SSIM and ORB feature comparison. Thus, adaptive confidence fusion allows to make a better forgery detection than using only semantic or structural cues.

C. Qualitative Analysis

Figure 3 displays some typical samples that have been detected using our ReScene framework. Real revisit videos maintain both the semantic relation and structural consistency with the reference frame, leading to a low level of forgery confidence. On the other hand, the forged videos, which are made using object-level manipulations, have been flagged as inconsistent context by the MemoryGCN detector; the structural validation module detects any visual inconsistencies using SSIM and ORB methods.

D. Discussion

Unlike conventional approaches that rely primarily on pixel-level artifacts, *ReScene* integrates semantic scene graph reasoning and structure verification in order to find both context and image inconsistency in revisit scenes. While ReScene is tested on the proposed revisit video dataset, in future research the framework may be applied to a much larger data set and different manipulation settings.

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